

Taming diverse metadata from heterogeneous bioactivity assays

Noel O'Boyle

Chemical Biology Resources Team Leader

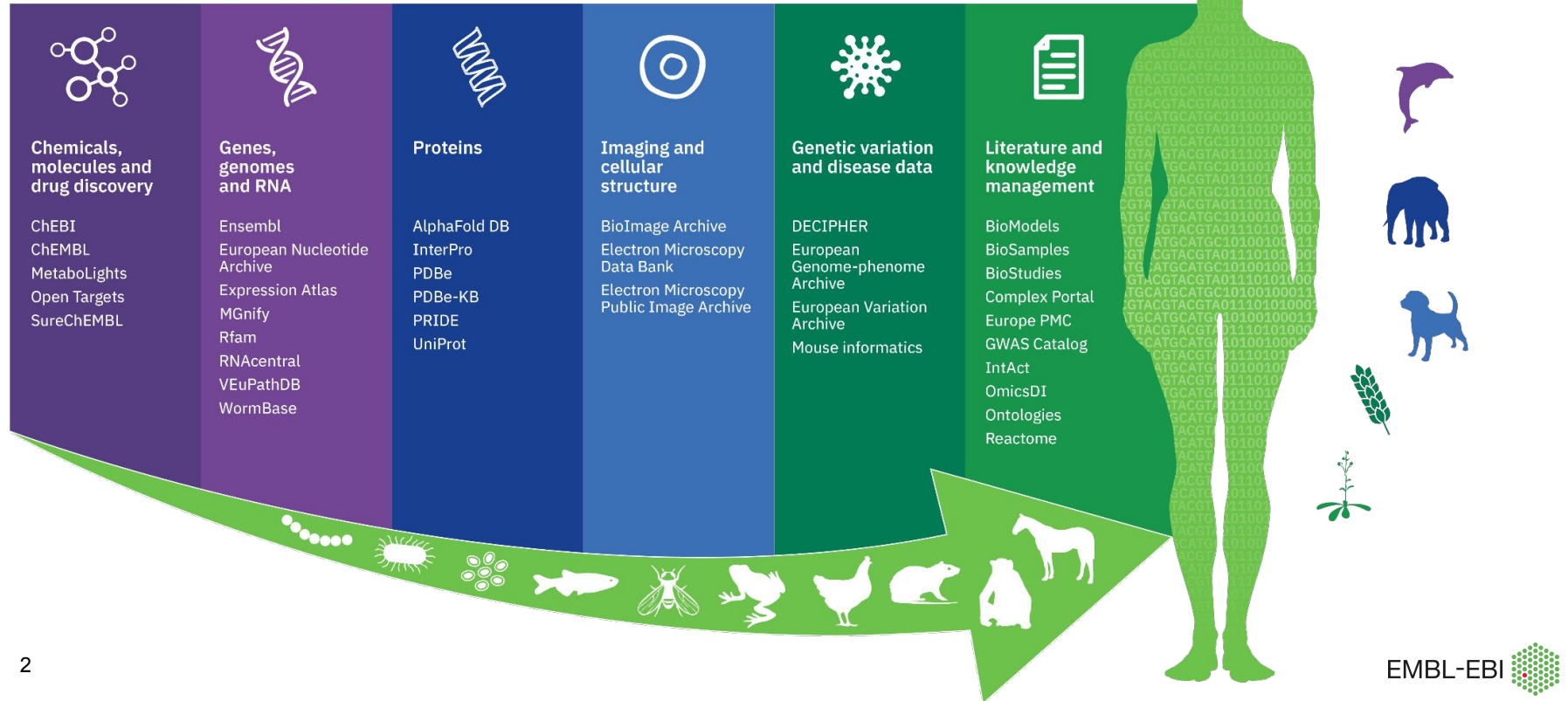
Emma Manners, Barbara Zdrazil

Senior Biological Curator, ChEMBL Team Coordinator

ACS Fall
26/07/2026



Data resources at EMBL-EBI (Hinxton, UK)



Chemical Biology Resources @ EMBL-EBI



ChEMBL

*Bioactivities from
literature &
deposited*

*Drug/candidate
information*



SureChEMBL

*Structures from
patents*

Patent annotations



UniChem

*Link databases via
InChI*



ChEBI

Chemical ontology

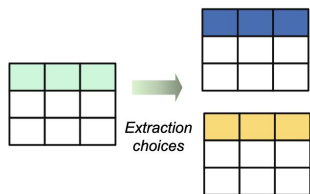
*Chemical registration
db for biological dbs*



OPSIN

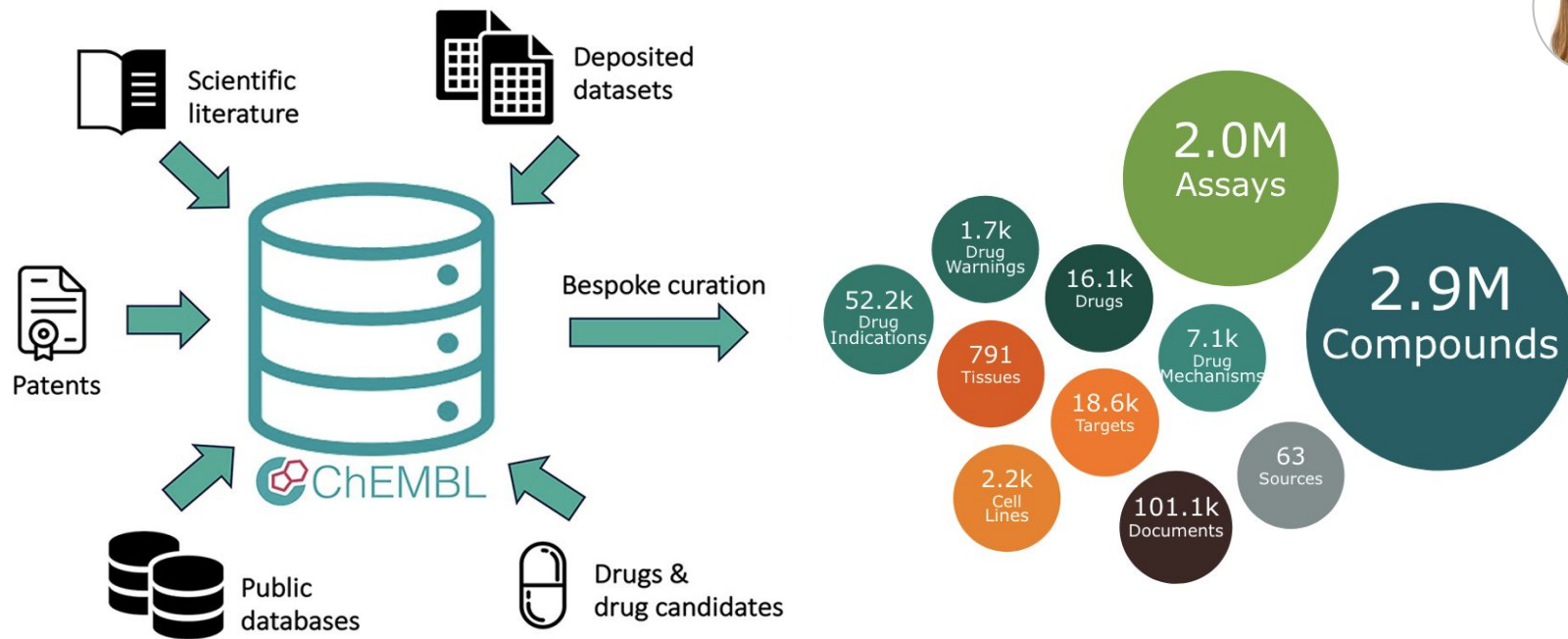
*Convert IUPAC
names to structure*

Overview



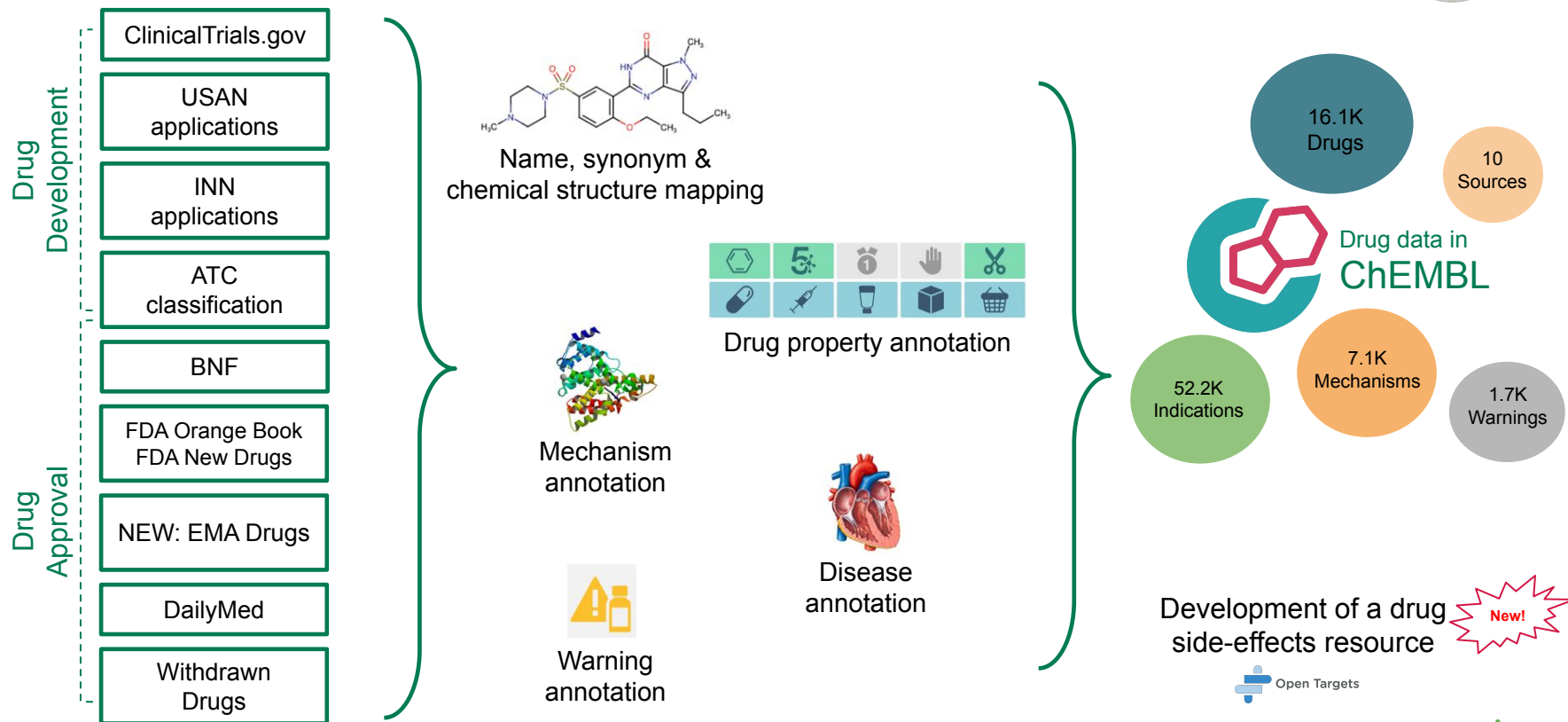
1. Background to the ChEMBL database
2. Data heterogeneity: what users need to understand and check, and how we can help
3. Dataset construction: why extraction choices matter

Bioactivity data for drugs and druglike compounds

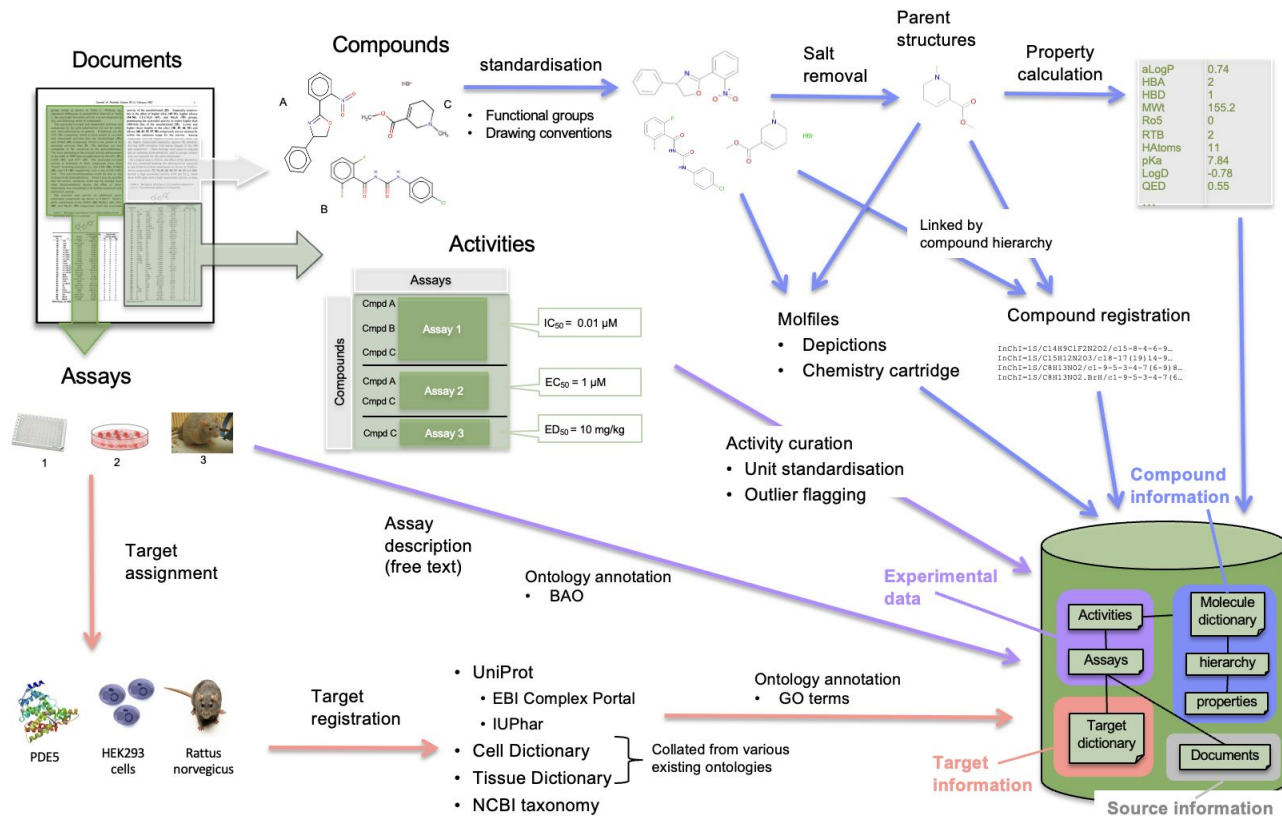


Diverse data types; broad coverage of compounds & targets; FAIR and curated

Extensively curated drug & clinical candidates



Expert curation and standardisation



Bento, A.P. et al. **An open source chemical structure curation pipeline using RDKit.** *J. Cheminform.* **2020**, *12*, 51.

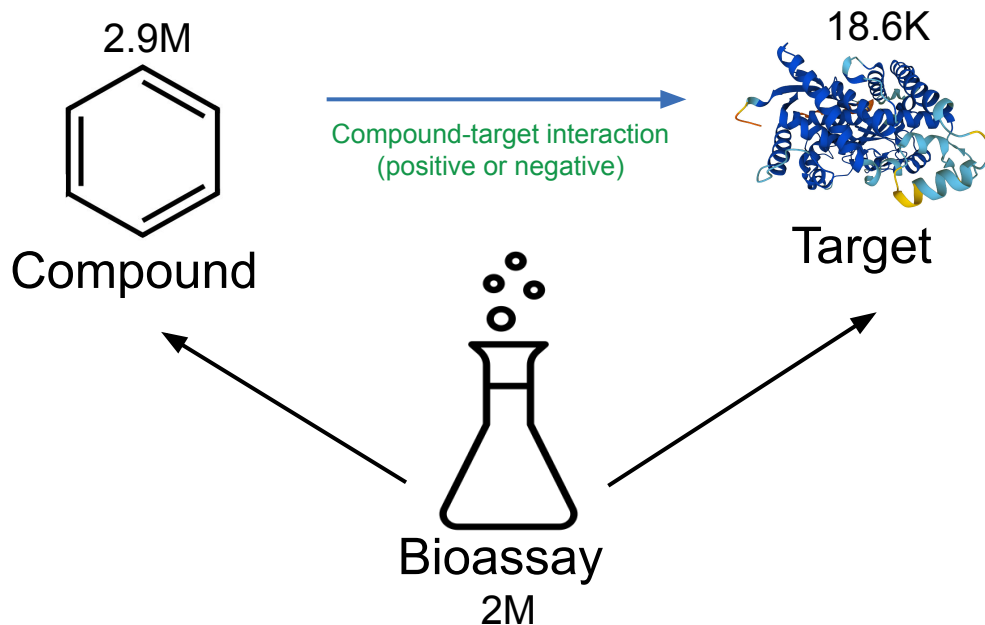
https://github.com/chembl/ChEMBL_Structure_Pipeline

Papadatos G. et al. **Activity, assay and target data curation and quality in the ChEMBL database.** *J. Comput. Aided Mol. Des.* **2015**, *29*, 885.

ChEMBL drives bioactive compound research



Provides bioactivity **data** and **metadata** describing **compound-target interactions** that underpin drug discovery.



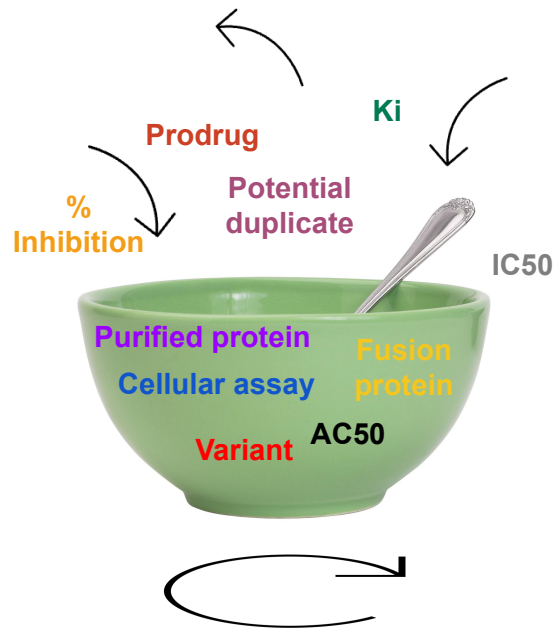
Wordcloud produced from articles between 2010 and 2024 (n = 912) with "ChEMBL" in title or abstract.

(Some) Applications
Target identification and validation
Polypharmacology & off target analyses
QSAR models
ADME, toxicity, and physicochemical studies

The challenge: heterogeneous data



ChEMBL
team

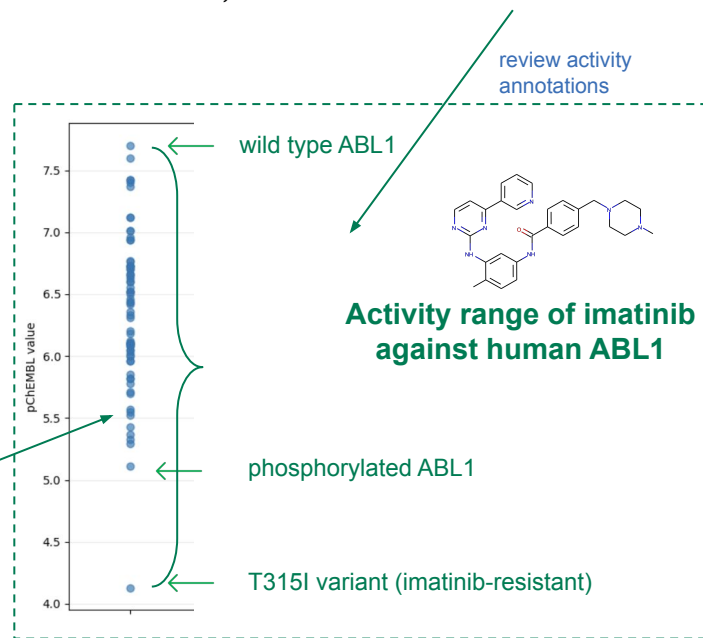


ChEMBL users



*"Is public bioactivity data
low quality?"*

*"Data reflects the source -
design queries, review,
filter, and check data"*



Understand the data features that shape output

1 Protein variants alter bioactivity profiles

Excuse me, there is a mutant in my bioactivity soup!

A comprehensive analysis of the genetic variability landscape of bioactivity databases and its effect on activity modelling

Marina Gorostiola González^{1,2,‡}, Olivier J. M. Béquignon^{1,‡}, Emma J. Manners³, Anna Gaulton³, Prudence Mutowo³, Elisabeth Dawson³, Barbara Zdrazil³, Andrew R. Leach³, Adriaan P. IJzerman¹, Laura H. Heitman^{1,2}, and Gerard J. P. van Westen^{*1}

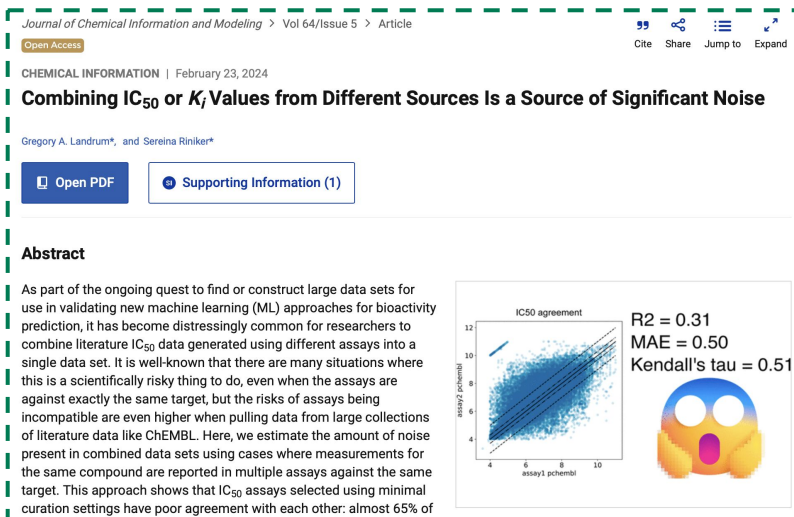
¹Leiden Academic Centre of Drug Research, Leiden University, 55 Einsteinweg, 2333 CC. Leiden, The Netherlands

²Onco Institute, The Netherlands

³European Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Wellcome Genome Campus, Hinxton, Cambridgeshire, CB10 1SD, United Kingdom

[‡]These authors contributed equally

^{*}Corresponding author (gerard@lacdr.leidenuniv.nl)



2 Combining disparate data may be problematic

Good data practices are essential - models improve with carefully selected and “cleaned up” input

What can users expect from ChEMBL?

- ✓ Extracted medicinal chemistry literature data disseminated in a structured form
- ✓ Curation and standardisation where appropriate
- ✓ Periodic enhancements and updates (when appropriate, and resource allows)
- ✓ Training and regular publications describing data content, structure, and curation
- ✓ Correction of reported errors in extraction



ChEMBL Deposition: Expectations and Promises

Curated data still requires user review

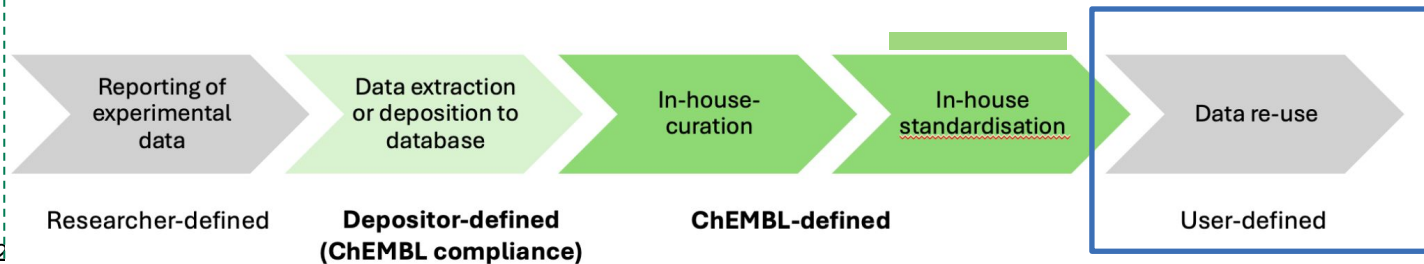
- ✗ We **do not** peer-review extracted literature and cannot guarantee error-free sources
- ✗ We **cannot avoid** some errors during data ingestion and integration
- ✗ We have **very limited control** over data reuse by our users

*Understanding potential errors, data **structure**, and relevant **annotations** reduces noise during data **reuse***

Sources of potential error/noise in bioactivity datasets

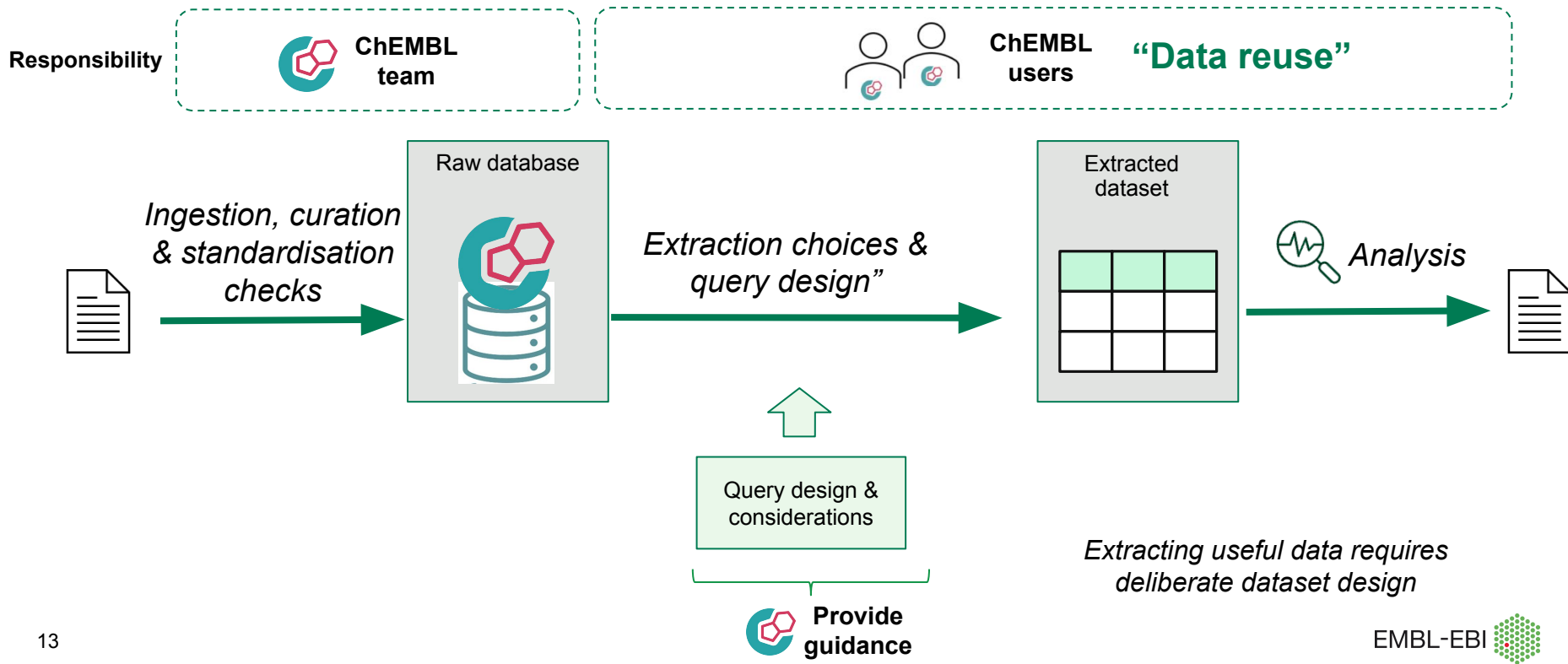
• Changes in data reporting over time • New annotations, entities, etc. • Challenges in (re)annotating legacy data

- Reporting errors
- Extraction errors
- Entity mapping errors
- Pipeline errors
- Data structure not considered
- Omissions
- Omissions
- Missing identifiers
- Limited data checks/clean-up

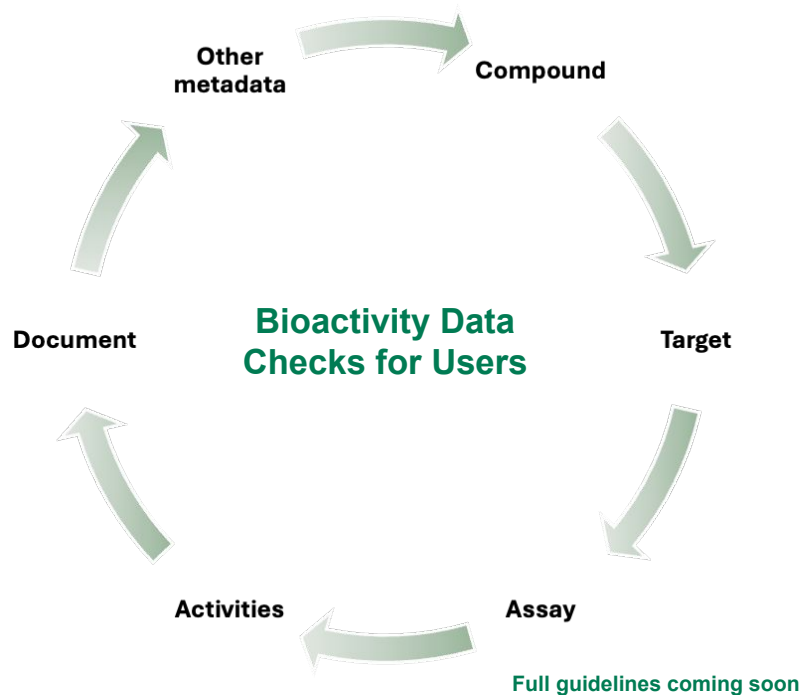


ChEMBL can help with dataset preparation

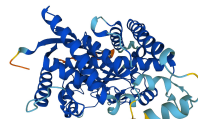
Raw database $\neq$ ML-ready, relevant dataset



Guidelines for *Bioactivity Data Checks for Users*



*Rules, annotations, and data structure should be considered for **each** entity type*



ANNOTATION FIELD (TABLE)
VARIANT_ID (ASSAYS)
VARIANT_SEQUENCES
ASSAY_STRAIN (ASSAYS)
CONFIDENCE_SCORE (ASSAYS)
HOMOLOGUE (ASSAYS)
TARGET_TYPE (TARGET_DICTIONARY)

Target/assay annotations

Compound data checks in practice

*Evaluating and curating **compound** features for model input*

Dataset design: **relevant** + **valid** data -> **dataset composition suited to goal**

Compound representation

- ☐ Compound type
- ☐ Single/multi-component molecule
- ☐ Structural representation
- ☐ Duplicates

Compound identity

- ☐ Identifiers
- ☐ Match to (curated) drug?
- ☐ Parent/child forms
- ☐ Mixture/racemate/stereoisomer
- ☐ Prodrug vs active metabolite

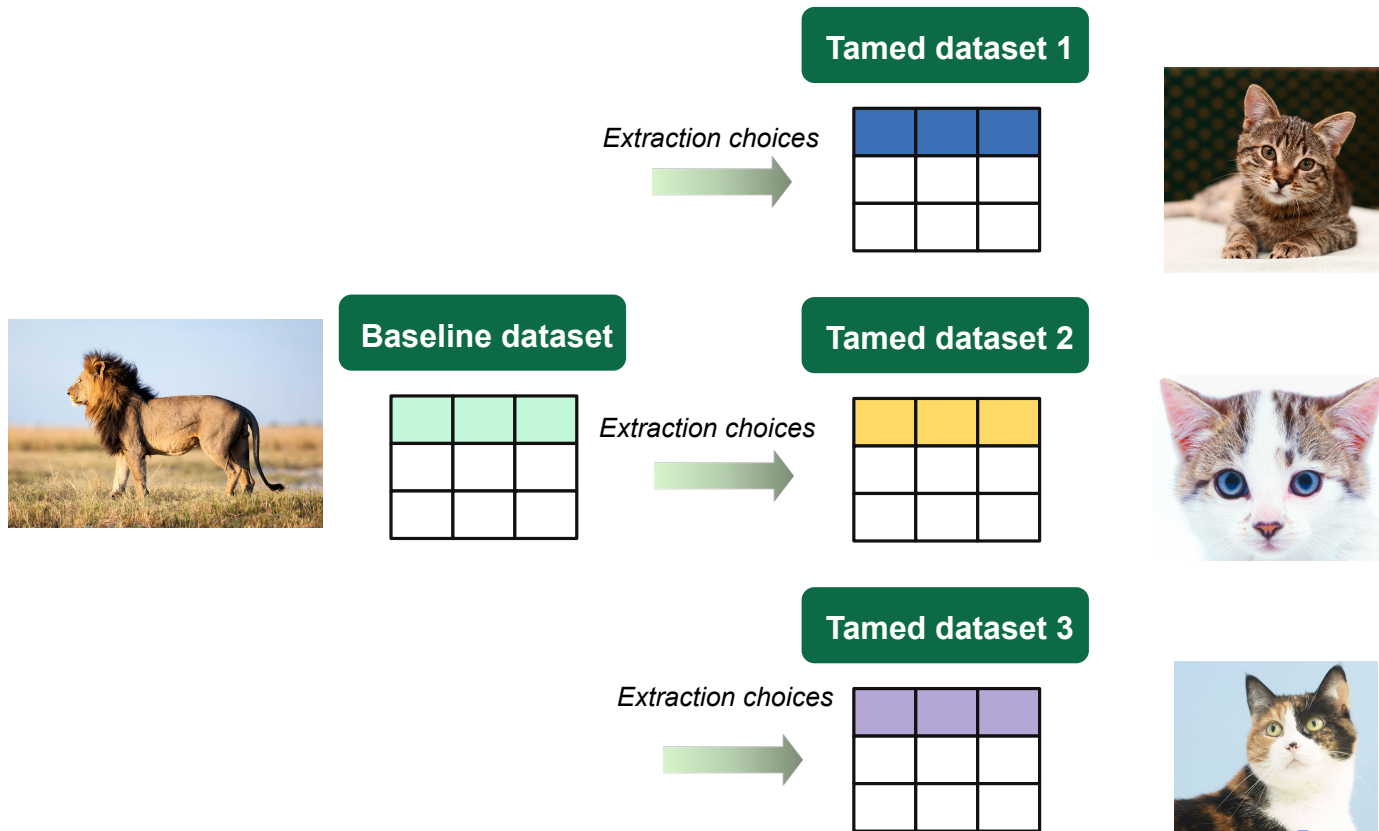
Compound form

- ☐ Tautomers / protonation state
- ☐ Stability - reaction products - structural alerts
- ☐ Prodrug - active form liberated in assay?



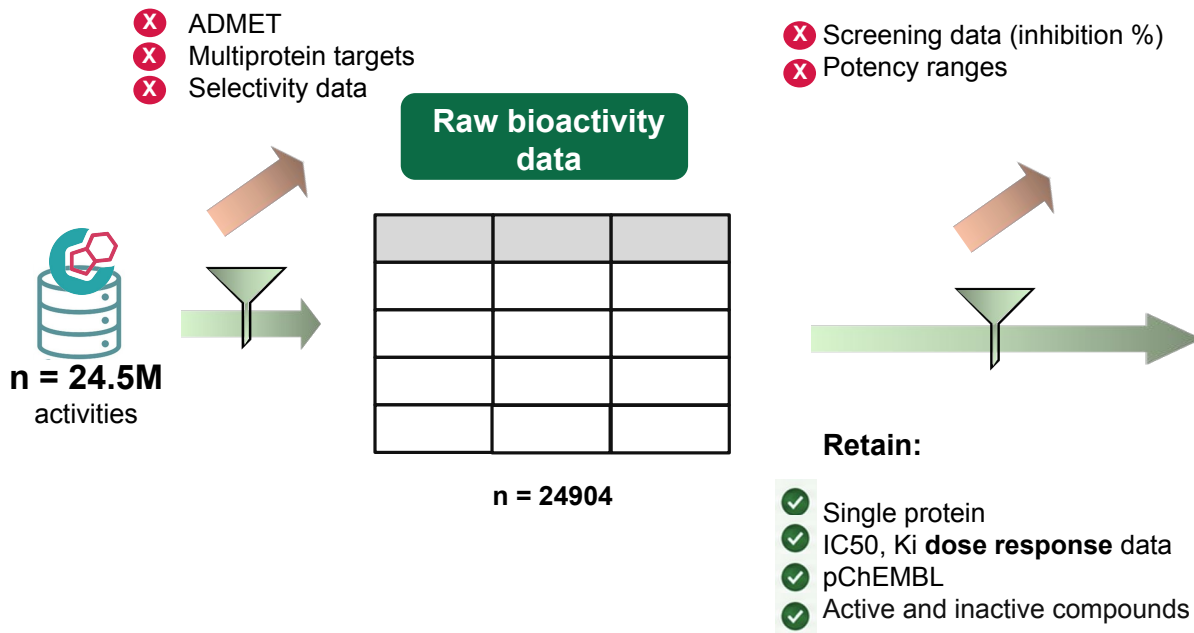
**Model
ready**

How to “tame” a dataset: examples



How is ChEMBL data typically extracted?

A typical process: extract bioactivities for human ABL1 kinase



Be aware of what you're losing

Baseline dataset

n = 5304
activities

- ✓ Model friendlier
- ? Reduced insight?
- ↓ Lost novelty?

Heavily depends on the research question

Constructing a baseline dataset

Taming: which parameters can be adjusted?

A typical process:

extract bioactivities for human ABL1 kinase

Baseline dataset

n = 5,304

- ✓ Single protein
- ✓ IC50, Ki
- ✓ pChEMBL



Tamed dataset 1

n = 4846

- Duplicates
- Data validity warnings



Tamed dataset 2

n = 4580

- Variants



Tamed dataset 3

n = 300

- ✓ Compounds with data from at least 3 documents



Tamed dataset 4

n = 1007

- ✓ Compounds with at least 5 activities



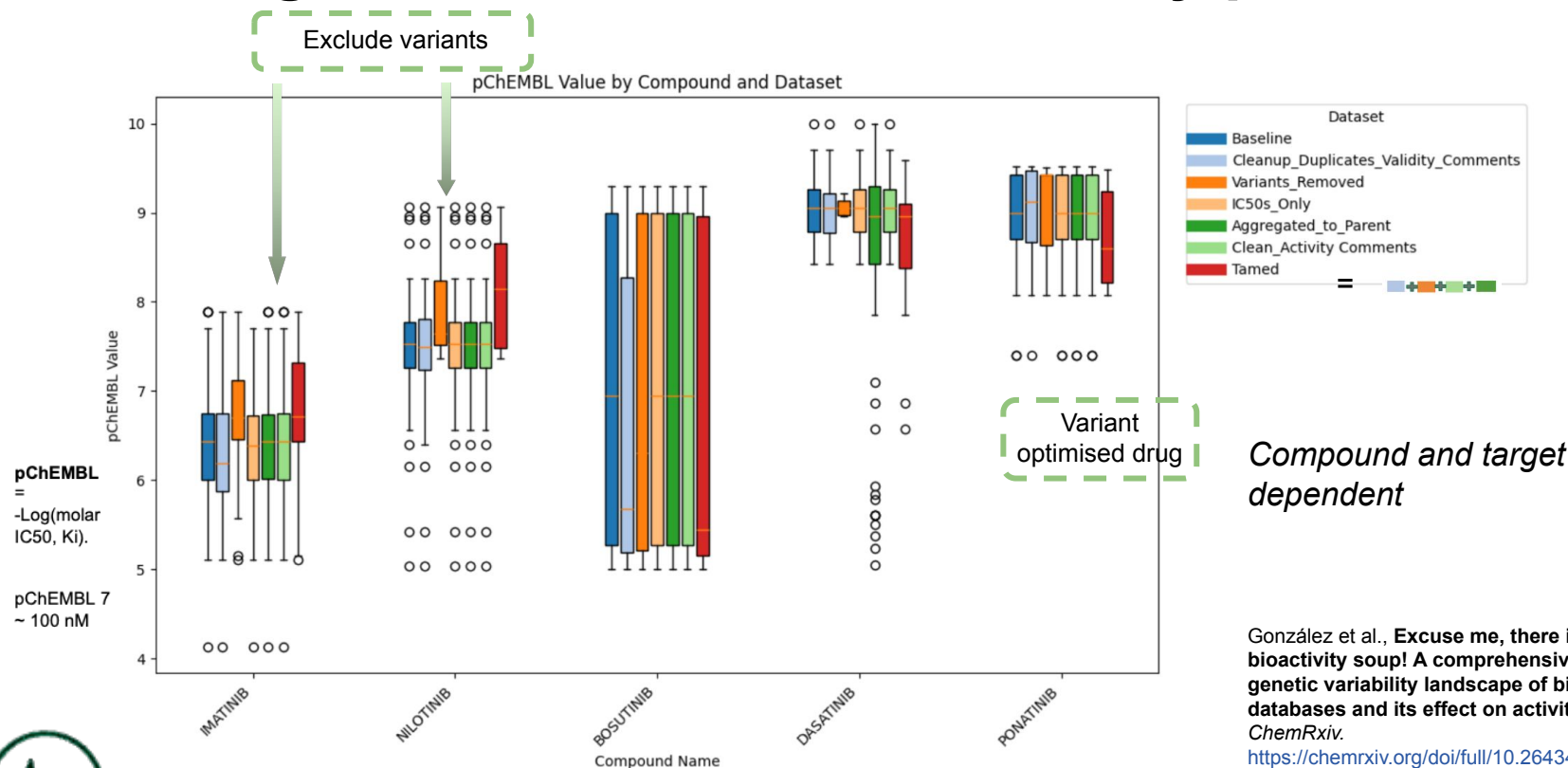
Refining the baseline dataset

Extraction choices produce controlled data over "big" data

EMBL-EBI



Excluding variants shifts ABL1 activity profiles



González et al., **Excuse me, there is a mutant in my bioactivity soup! A comprehensive analysis of the genetic variability landscape of bioactivity databases and its effect on activity modelling.** *ChemRxiv*.
<https://chemrxiv.org/doi/full/10.26434/chemrxiv-2024-kxlgm>



Top-ranked compounds change across *tamed* datasets

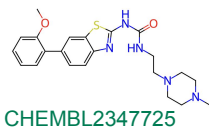
Compounds with the highest activity

Baseline dataset

Wild-type ABL1

Variant ABL1

Variant ABL1
T315I tested



1. CHEMBL2347725 None	1. CHEMBL3689722 None	1. CHEMBL2347725 None
2. CHEMBL3689722 None	2. CHEMBL1221411 None	2. CHEMBL2347722 None
3. CHEMBL1221411 None	3. CHEMBL3689678 None	3. CHEMBL2347711 None
4. CHEMBL3689678 None	4. CHEMBL3689631 None	4. CHEMBL2347723 None
5. CHEMBL3689631 None	5. CHEMBL3689554 None	5. CHEMBL2347710 None
6. CHEMBL3689554 None	6. CHEMBL3685180 None	6. CHEMBL2347719 None
7. CHEMBL3685180 None	7. CHEMBL3685101 None	7. CHEMBL2347729 None
8. CHEMBL3685101 None	8. CHEMBL3685056 None	8. CHEMBL2347728 None
9. CHEMBL3685056 None	9. CHEMBL3639840 None	9. CHEMBL2347724 None
10. CHEMBL3639840 None	10. CHEMBL2347711 None	10. CHEMBL2347727 None

Retained from baseline Lost from at least one refined list New versus baseline

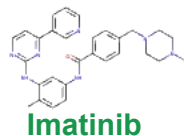
- Query design changes the most potent compounds - those that may be taken forwards.

- Filtering in/out *variant* ABL1 delivers very different list of top-ranked compounds.

Change in most potent ABL1 kinase inhibitors in *tamed* datasets

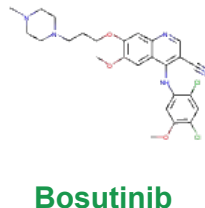
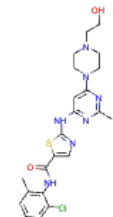
Biological basis for altered variant ABL1 activity

1st gen

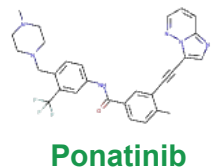


Approved drugs that
bind ABL1 kinase

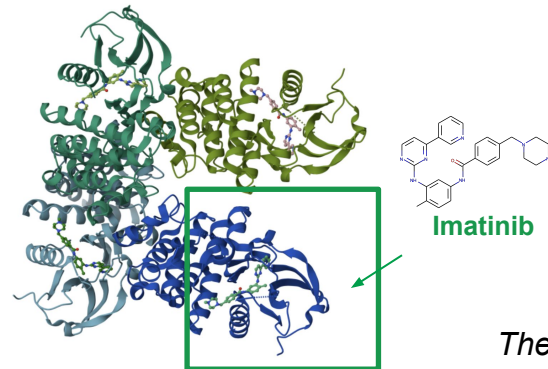
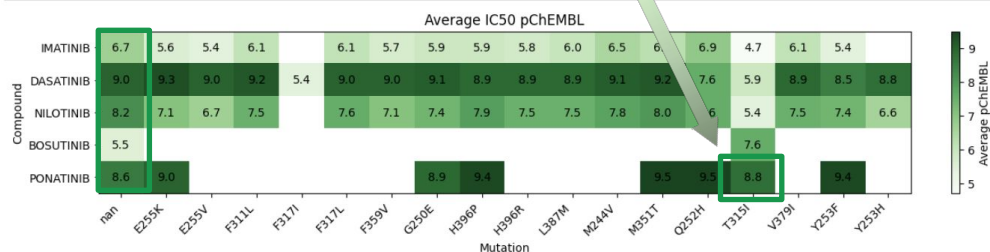
2nd gen



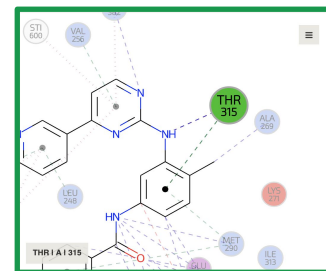
3rd gen



*Ponatinib overcomes
T315I drug-resistance*



ABL kinase + Imatinib (PDB; 2hyy)



The ABL1 kinase
T315 residue
forms H-bonds
with imatinib, this
binding
interaction is
disrupted in the
T315I
drug-resistant
variant

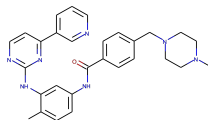
Single protein datasets can miss disease-relevant multiproteins

Single proteins provide high confidence datasets, but may miss disease-relevant associations

BCR

ABL1

Oncogenic BCR-ABL1 fusion causes acute myeloid leukemia - locks ABL1 in an "active" (open) conformation.



Imatinib
IC₅₀ = 2.79 nM
[Assay ChEMBL5115322]

Most active
against
ABL1

Compounds with highest activity

ABL dataset

BCR-ABL dataset

0 matches | 20 ABL only | 20 BCR-ABL only

1. CHEMBL2347725

None

2. CHEMBL3689722

None

3. CHEMBL1221411

None

4. CHEMBL3689678

None

5. CHEMBL3689631

None

6. CHEMBL3689554

None

7. CHEMBL3685180

None

8. CHEMBL3685101

None

9. CHEMBL3685056

None

10. CHEMBL3639840

None

1. CHEMBL6159841

None

2. CHEMBL334364

None

3. CHEMBL128706

None

4. CHEMBL6146115

None

5. CHEMBL6114872

None

6. CHEMBL6134515

None

7. CHEMBL6150452

None

8. CHEMBL6167998

None

9. CHEMBL2178270

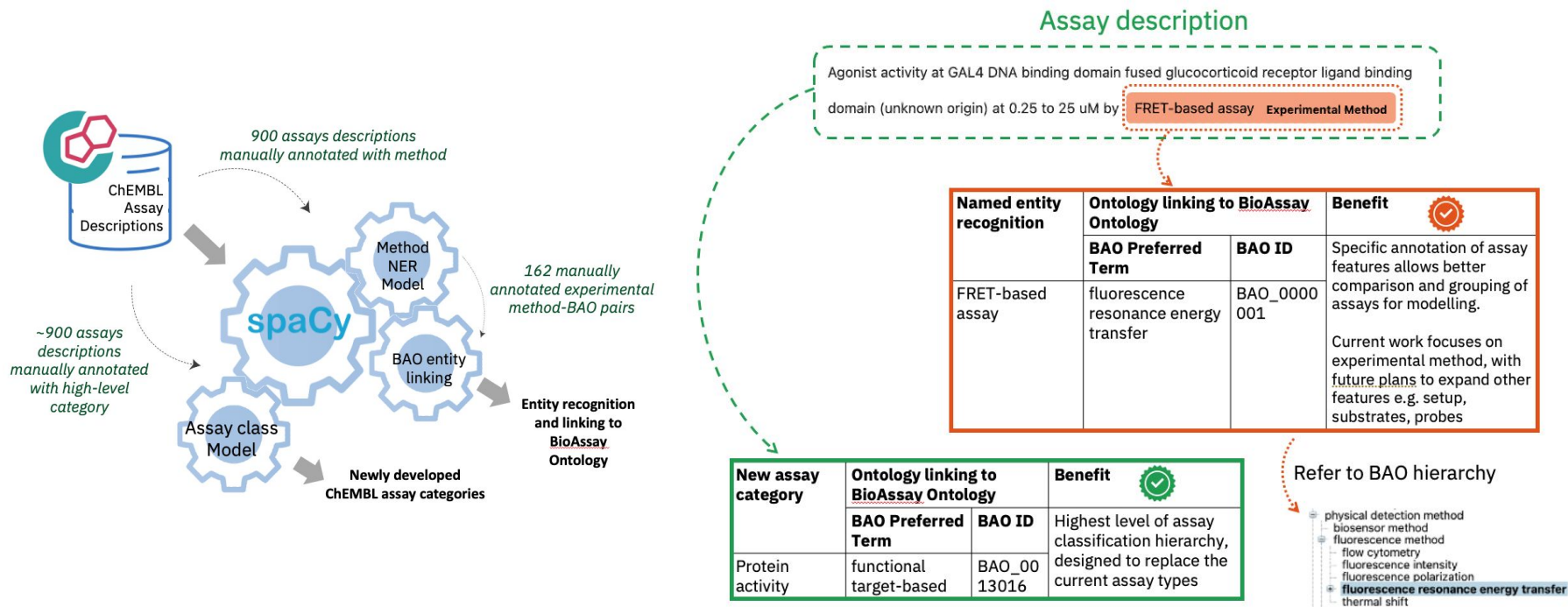
None

10. CHEMBL6148910

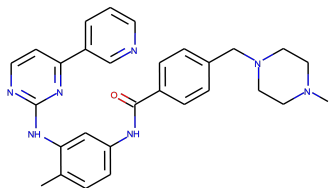
None

Most active
against
ABL1-BCR
fusion

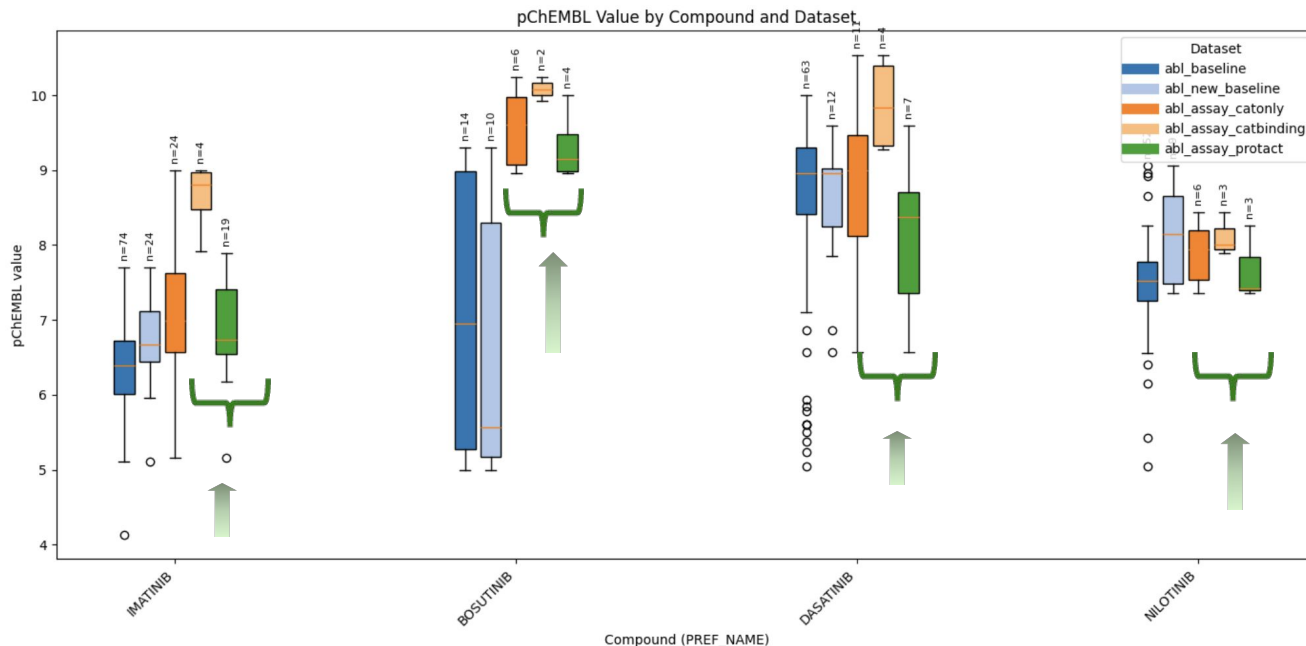
Enhanced assay annotations provide context



Assay category alters bioactivity profiles



**ABL1 kinase inhibitor:
imatinib (CHEMBL941)**



Splitting **binding** and **protein activity** assays consistently shifts median activity.

Data reuse “must haves”

- Knowledge of input data
- Clearly understood limitations
- Tailored data extraction process that defines all parameters:

Compound



degrader



antibody-drug
conjugate

Action type

Agonist EC50

Inhibitor IC50

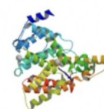
Degrader DC50

Target type/family

Protein-Protein
Interaction



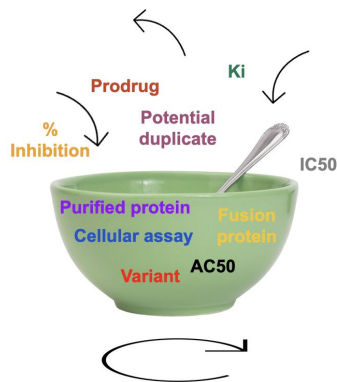
Single Protein



Data source



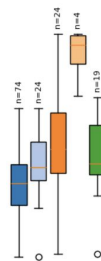
Take home messages



Bioactivity data are heterogenous by nature



Query design and data analysis are equally important



Data extraction choices impact results

Checklist



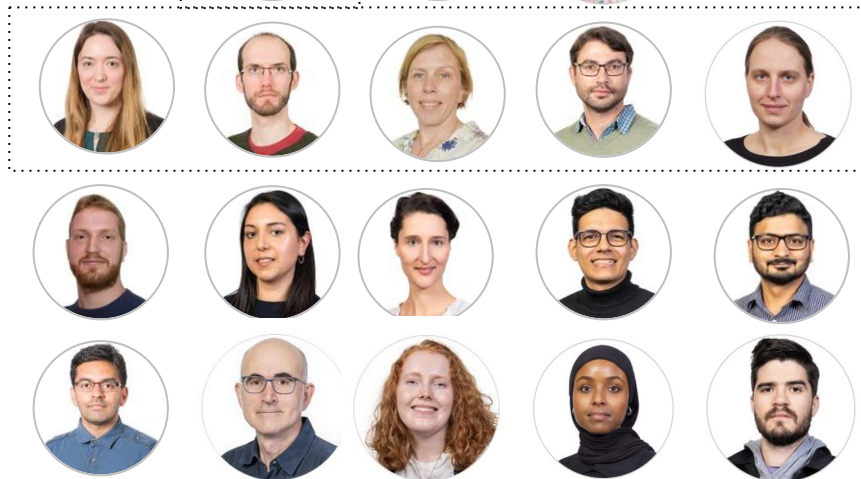
ChEMBL's data extraction guidelines coming soon!

Chemical Biology Resources

ChEMBL Coordinator



Data team



wellcometrust
Strategic Award



Questions about ChEMBL data? Corrections?
Email: chembl-help@ebi.ac.uk

Join the team!

- ChEBI chemistry curator (by Sept 9th 2026)
 - ARISE2 Fellowship opportunity (with AZ, get in touch ASAP by early Sept)
-Coming soon, biological curator, possibly also scientific developer

